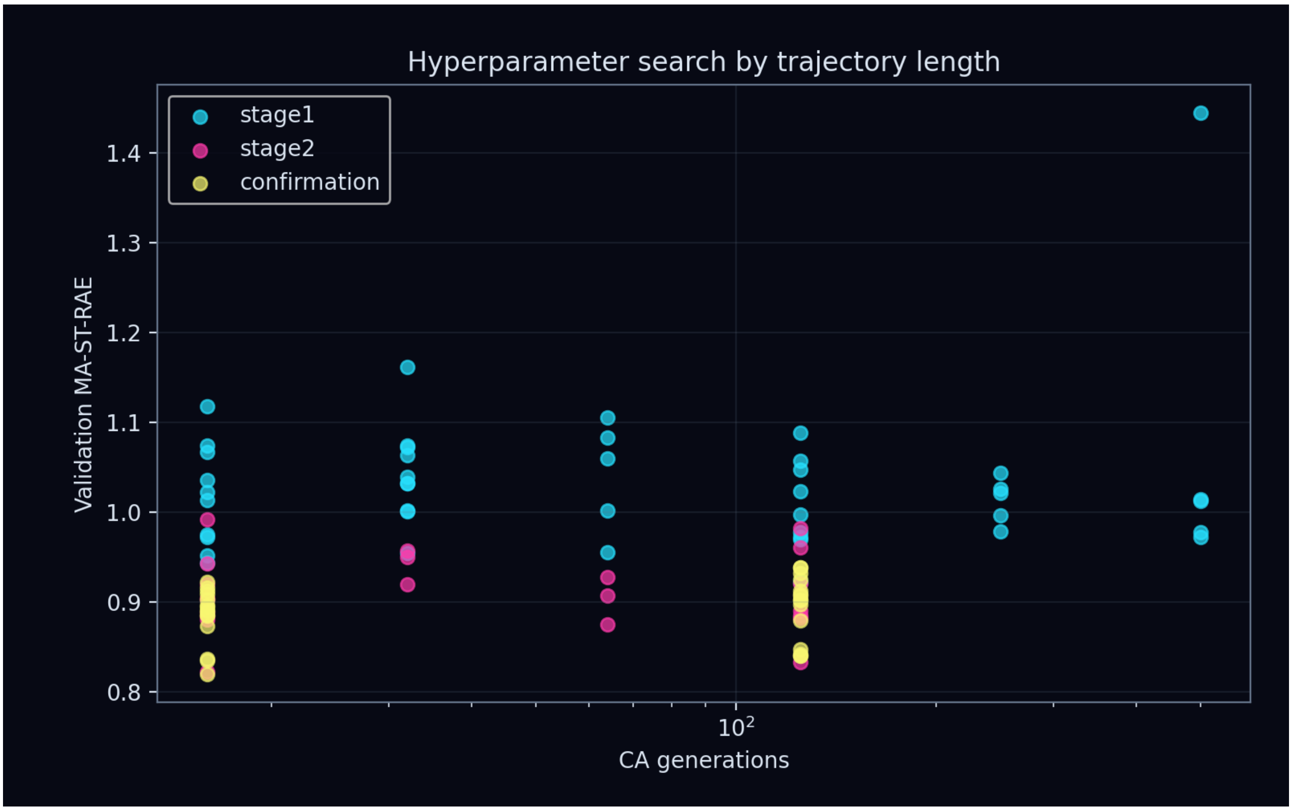
Strange Matter Engine

Production study: fitzhugh_nagumo

Direct-inhibition pIC50 | grouped validation | blinded set excluded

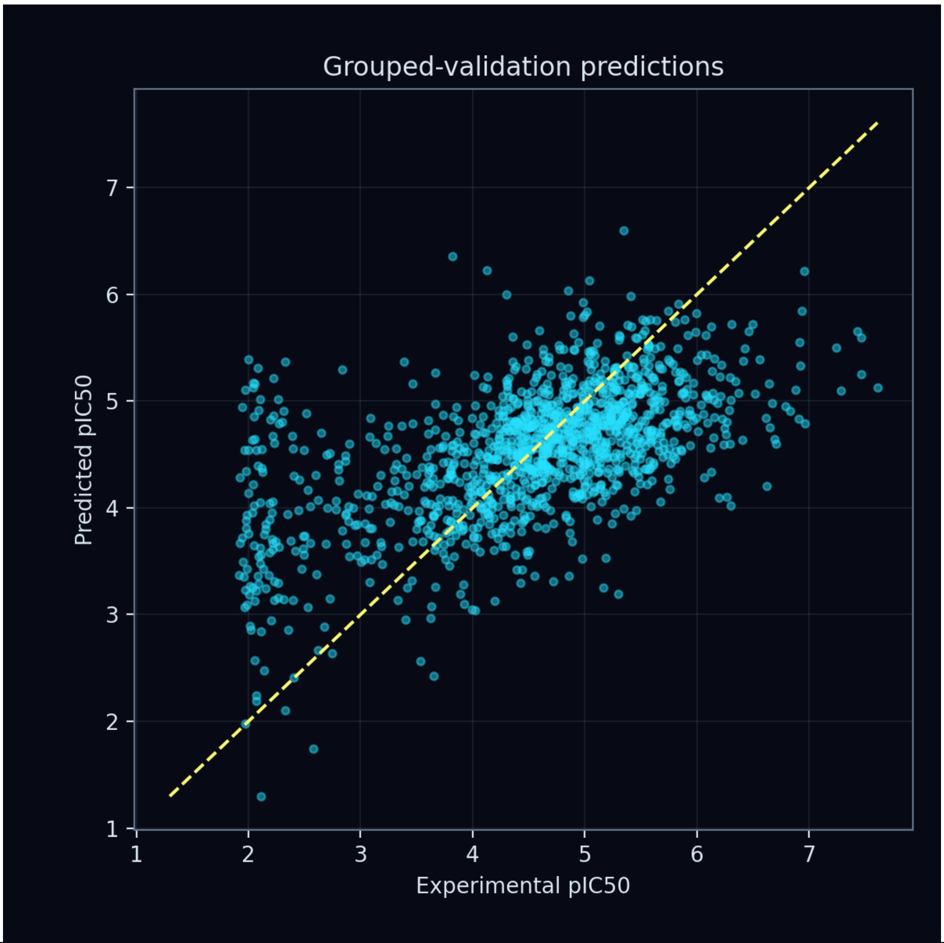
Quantity	Result
Validation MA-ST-RAE	0.8215
Fit MA-ST-RAE	0.8374
Validation RMSE	0.8819 pIC50
Fit RMSE	0.8283 pIC50
Generations	16
Dynamical channels	16
Atom feature profile	electronic
CA learning rate	0.001
Ridge penalty	0.1
CA L2	0.0001
Gradient clip	0.5
Confirmation mean	0.8845
Confirmation seed SD	0.0300
Bootstrap MA-MAE	0.6536 [0.6183, 0.6854]
Bootstrap MA-R2	0.2310 [0.1791, 0.2810]
Bootstrap MA-Spearman rho	0.4898 [0.4443, 0.5352]
Bootstrap MA-Kendall tau	0.3470 [0.3135, 0.3801]

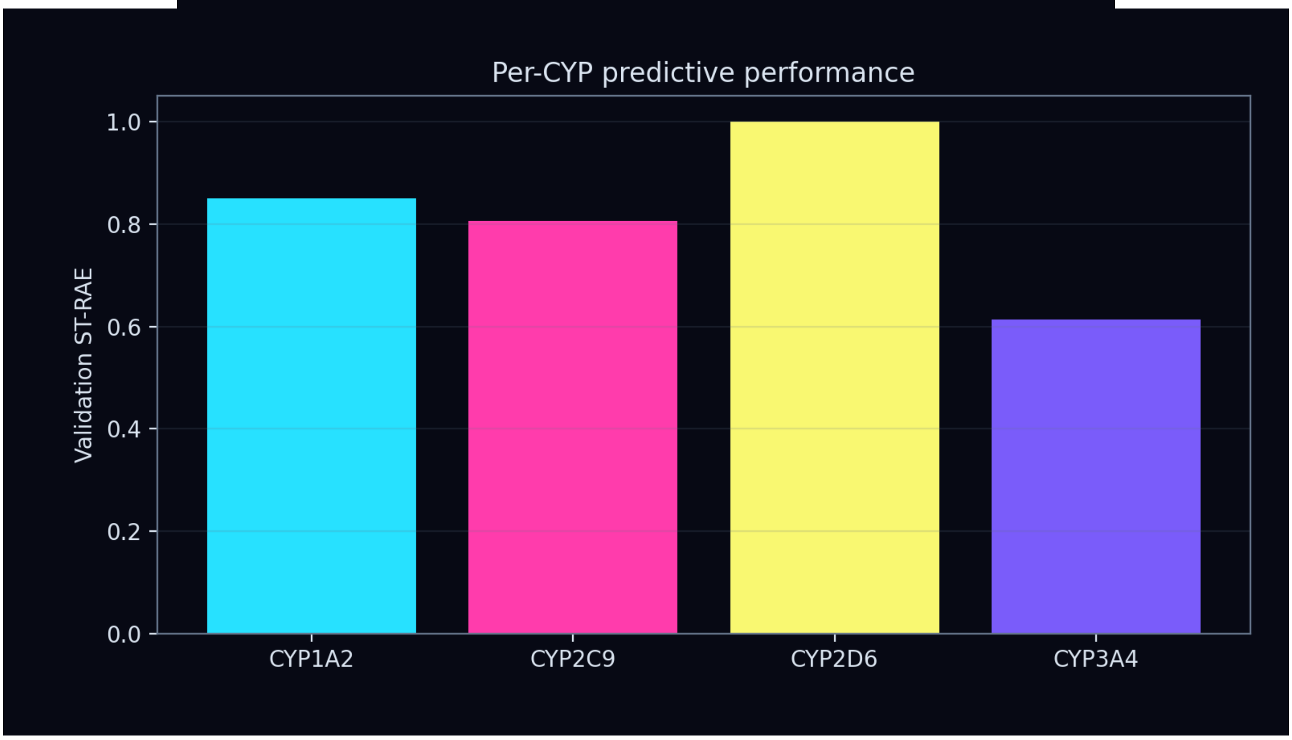
Enhanced CA controls: update scale 0.08, initial-state scale 1.0, support fraction 0.75, bond temperature 0.5. Rule dynamics: A=0.5, B=0.5, C=0.2, D=0.05.


Hyperparameter search by trajectory length
stage1
stage2
confirmation
Validation MA-ST-RAE
1.4
1.3
1.2
1.1
1.0
0.9
0.8
10^2
CA generations

Predictive validation

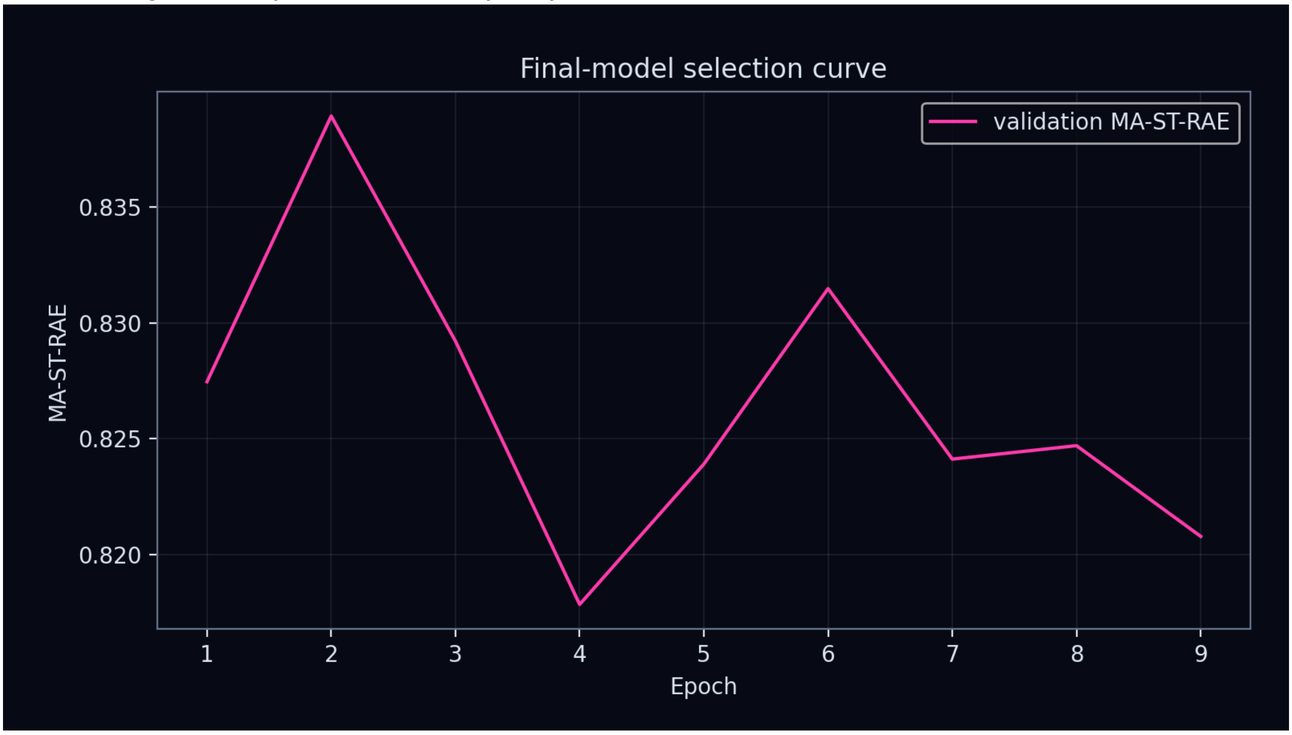
Model selection used grouped-validation MA-ST-RAE, the challenge primary metric. Dynamical-interest scores did not influence promotion.





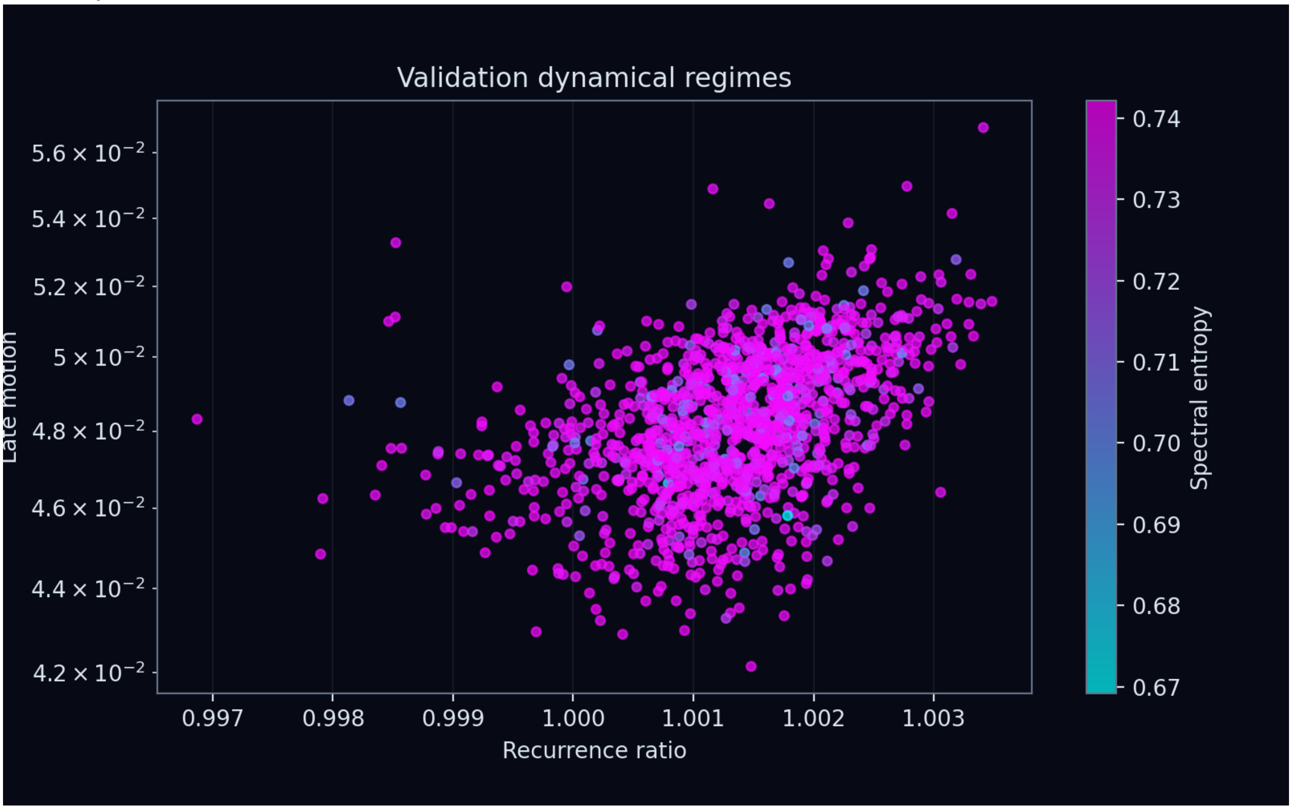
Optimization

Ridge coefficients and the unpenalized intercept were solved from permitted fitting fingerprints. Query error differentiated through the solve into the graph CA; Adam updated only CA parameters. Bond-conditioned messages, cosine learning-rate decay, and multitime trajectory statistics were shared across all ten rules.



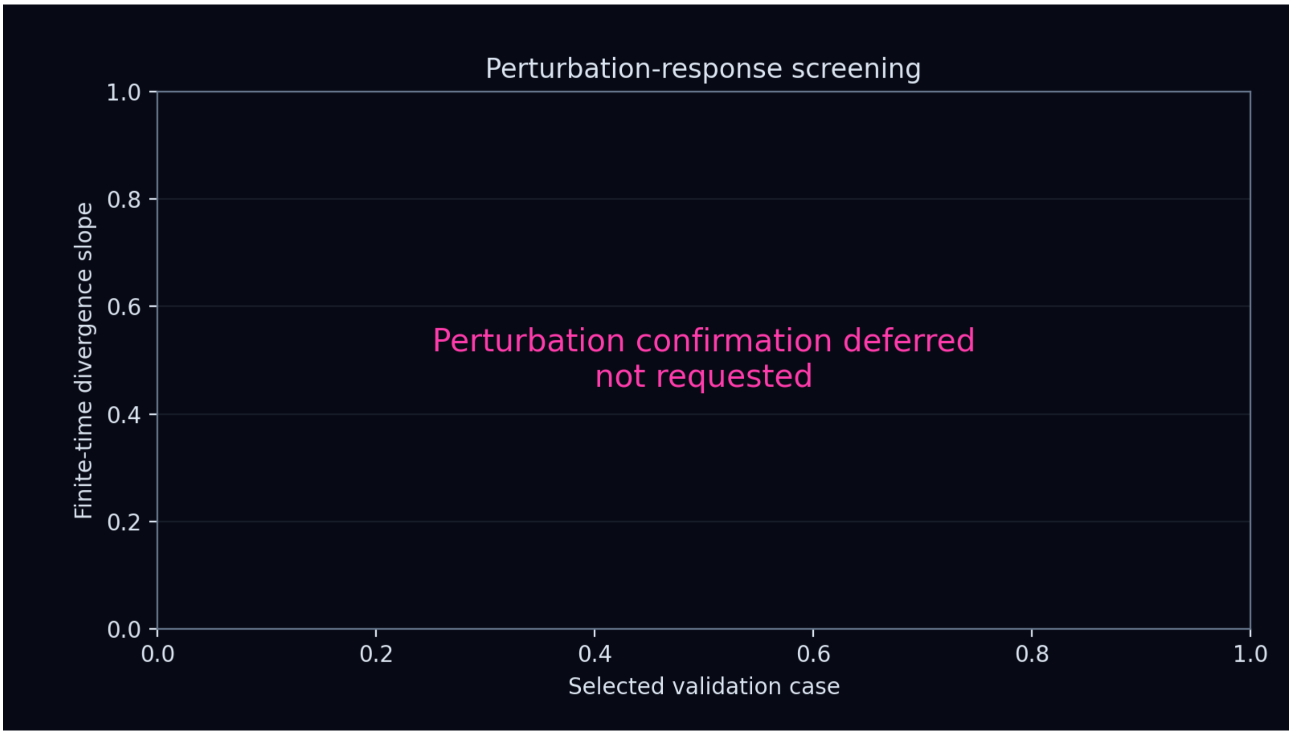
Emergent dynamics

The validation trajectories were screened for convergence, recurrence, persistent motion, spectral complexity, and perturbation sensitivity. Labels ending in candidate require longer, renormalized and seed-repeated confirmation before any attractor or chaos claim.



Permutation and perturbation screening

This figure tests whether small state perturbations separate or contract over the observed finite trajectory. It is placed on a dedicated page to preserve its axes, labels, and complete plotting area.



Scientific limitations

This report describes one transition-rule study under the declared grouped split. Hyperparameter selection and early stopping used labelled training data only. The blinded challenge set was not loaded or predicted. Finite-time positive slopes are screening signals rather than proof of a strange attractor or sustained chaos.